Machine Learning Laboratory

Experiment No. 9

Comparison of SVM Kernels on Fashion MNIST

Aim

Implement and compare the performance of SVM classifiers with Linear, Polynomial, and RBF kernels on the Fashion MNIST dataset. Analyze the advantages and disadvantages of each kernel type.

Objectives

- Load and preprocess the Fashion MNIST dataset.
- Implement SVM with Linear, Polynomial, and RBF kernels.
- Compare the classification performance for each kernel.
- Discuss the strengths and weaknesses of each kernel type.

Theory

Support Vector Machine (SVM) is a supervised machine learning algorithm used for classification and regression. It finds an optimal decision boundary that separates different classes while maximizing the margin between them.

For a linear SVM, the decision boundary is represented as:

$$w^T x + b = 0$$

where w represents the weight vector, x represents the input features, and b represents the bias.

SVM can use different kernel functions to handle linear and nonlinear relationships in the data.

Linear Kernel

The Linear kernel is given by:

$$K(x_i, x_j) = x_i^T x_j$$

It creates a linear decision boundary between the classes.

Polynomial Kernel

The Polynomial kernel is given by:

$$K(x_i, x_j) = (\gamma x_i^T x_j + r)^d$$

where d is the degree of the polynomial.

It can represent nonlinear relationships by mapping the data into a higher-dimensional feature space.

RBF Kernel

The Radial Basis Function (RBF) kernel is given by:

$$K(x_i, x_j) = \exp\left(-\gamma \|x_i - x_j\|^2\right)$$

The RBF kernel is useful for complex nonlinear decision boundaries.

Algorithm

Step 1: Start.

Step 2: Import the required libraries such as NumPy, Pandas, Matplotlib, and Scikit-learn modules.

Step 3: Load the Fashion MNIST dataset.

Step 4: Preprocess the dataset by separating the input features X and target labels y.

Step 5: Normalize the pixel values of the images.

Step 6: Split the dataset into training and testing sets.

Step 7: Create an SVM classifier using a Linear kernel.

Step 8: Create an SVM classifier using a Polynomial kernel.

Step 9: Create an SVM classifier using an RBF kernel.

Step 10: Train the Linear SVM classifier using the training data.

Step 11: Train the Polynomial SVM classifier using the training data.

Step 12: Train the RBF SVM classifier using the training data.

Step 13: Use the trained Linear, Polynomial, and RBF SVM classifiers to predict the class labels of the test data.

Step 14: Calculate the classification accuracy of each classifier.

Step 15: Compare the performance of the Linear, Polynomial, and RBF SVM classifiers based on their accuracy.

Step 16: Display the accuracy results using a suitable table or graph.

Step 17: Analyze the advantages and disadvantages of each kernel based on classification performance, computational complexity, and ability to handle nonlinear relationships.

Step 18: Stop.

Performance Metric

The classification accuracy is calculated as:

$$\text{Accuracy} = \frac{\text{Number of Correct Predictions}}{\text{Total Number of Predictions}} \times 100$$

The performance of the three SVM classifiers is compared using:

$$\text{Accuracy}_{Linear}$$

$$\text{Accuracy}_{Polynomial}$$

$$\text{Accuracy}_{RBF}$$

Advantages and Disadvantages

Linear Kernel

Advantages:

- Simple and computationally efficient.
- Suitable for high-dimensional data.
- Requires fewer parameters.

Disadvantages:

- Cannot effectively model complex nonlinear relationships.
- May provide lower accuracy when a nonlinear decision boundary is required.

Polynomial Kernel

Advantages:

- Can model nonlinear relationships.
- The degree of the polynomial controls the complexity of the decision boundary.

Disadvantages:

- More computationally expensive than the Linear kernel.
- Performance depends on parameters such as C, γ, and polynomial degree.
- A high polynomial degree may lead to overfitting.

RBF Kernel

Advantages:

- Can model complex nonlinear decision boundaries.
- Suitable for datasets with nonlinear relationships.
- Often provides good classification performance.

Disadvantages:

- Computationally more expensive than the Linear kernel.
- Requires proper selection of C and γ.
- Training can be slow for large datasets.

Expected Outcome

- SVM classifiers with Linear, Polynomial, and RBF kernels successfully classify the Fashion MNIST dataset.
- The classification performance of the three kernels is compared using accuracy.
- The Linear kernel provides a simple and efficient classifier but may not capture complex nonlinear relationships.
- The Polynomial kernel captures nonlinear relationships, with its performance depending on the selected polynomial degree and other parameters.
- The RBF kernel captures complex nonlinear relationships but requires more computational time and proper parameter selection.

Result

SVM classifiers with Linear, Polynomial, and RBF kernels were implemented on the Fashion MNIST dataset. The classification performance of the three kernels was compared using accuracy. The strengths and weaknesses of each kernel were also analyzed.